Power BI Service

End-to-End Guide

Step by step, from account setup to dashboards, sharing, row-level security and administration

Project: Maven Inspectional Services (restaurant health inspections)

Built from: Up & Running with Power BI Service (Maven Analytics, 131-page deck) and your file: Powerbi_service_first_project.pbix

Compiled October 14, 2026

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How to use this guide

This guide walks through the whole Power BI Service project in the order you would actually do it. Every section says which slides of the course deck it comes from ("Deck p.NN"), so you can cross-check any step against the original.

What the guide is built from

- **Course deck** (Up & Running with Power BI Service, Aaron Parry, Maven Analytics): all 131 pages were read, including the screenshots, which carry many of the exact button names used below.
- **Your .pbix file** (Powerbi_service_first_project.pbix): inspected, see Appendix A for exactly what it contains.
- Steps are written as numbered lists. Button names, menu names and field names are in **bold** or `code font` exactly as the deck shows them.

Three honesty notes you should know

VERIFY

1. Screens change. The deck was recorded in 2024. Microsoft renames and moves buttons often (the deck itself warns about this on p.4). If a button is not where this guide says, look for the same wording nearby.

VERIFY

2. DAX formulas. The deck does NOT print the DAX for the measures in the semantic-model assignment (it says "see text lesson for details"). The formulas in this guide are suggested starting points built from the column names visible in the deck. Check them against your own column names and the course's text lesson. The only DAX shown in the deck itself is the row-level-security filters (Part 11).

VERIFY

3. Your .pbix is a starter file. It holds a single table called Violations (Demo) and one blank page. It does not hold the Maven Inspectional Services model, so the full project is built from the course data files, not from this file. Details in Appendix A.

What the colour boxes mean

Box	Meaning
IMPORTANT	A warning or rule from the deck that will break the project if ignored.
PRO TIP	A time-saving tip taken from the deck.
CHECK	A quick test to confirm the step worked.
EXTRA	A standard Power BI step that the deck does not spell out but you will need.
VERIFY	Something to double-check on your own screen.

What you need before you start

Item	Why	Deck
Work or school email address	Power BI needs one. @gmail, @yahoo, @outlook and similar are refused.	p.21
OR a free Maven Analytics account	Gives a free @mavenbi.org Power BI Service login if you have no work/school email.	p.22
Power BI Desktop (Windows)	Used to build the semantic model and publish it. Some features are not available on Mac.	p.4, 62
Course data files	Restaurant Inspection Data.csv plus the calendar, city, inspector, region, restaurant name and sanitary grade tables, and the Restaurant Inspections workbook.	p.35, 52
Windows PC that stays on	Needed for a data gateway if you load files stored on your computer.	p.37
OneDrive for Business	Required to upload a file or browse OneDrive when creating a dataflow.	p.45
Pro or PPU licence (or Premium/Fabric capacity)	Sharing, workspaces, apps and collaboration need it. The 60-day trial covers this.	p.4, 17, 102

Part 1. The project in one page

The course project is a case study. You have just been hired as an analyst for Maven Inspectional Services, a restaurant health inspection agency based in Los Angeles, California.

The brief

Source: Deck p.6

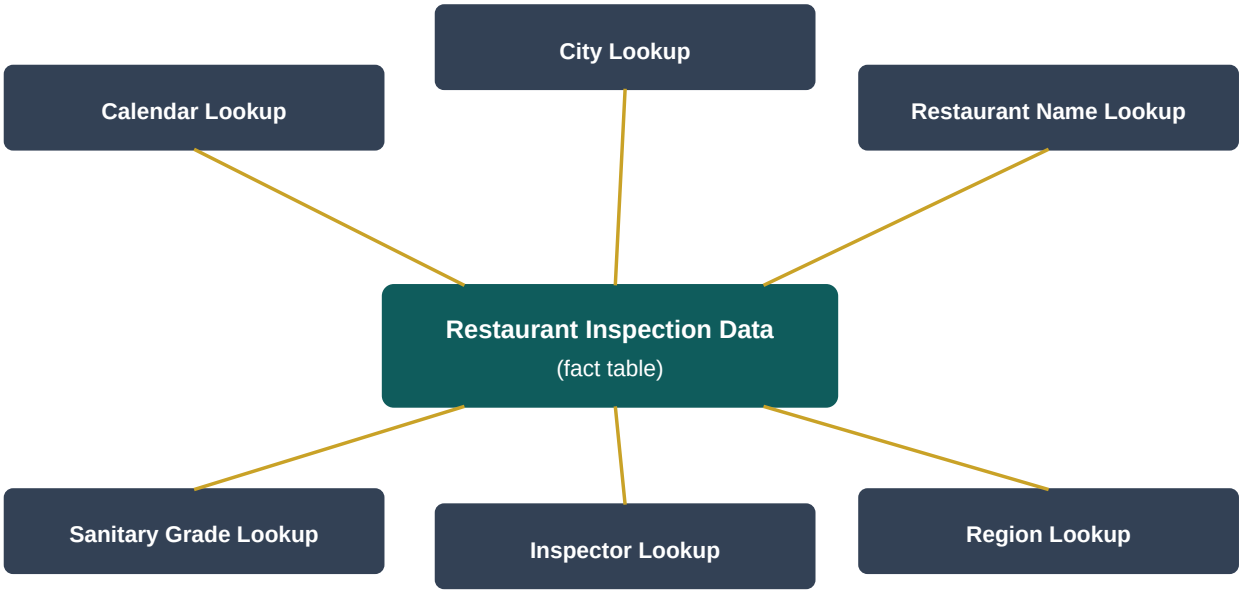
- **Situation:** all you have is a set of raw CSV files with health-inspection records plus details about each restaurant, inspector and violation.
- **Goal:** use Power BI Service (and Desktop) to build scalable reports and dashboards that can be shared across the company.
- **Objectives:** (1) load the raw data into Power BI Service, (2) build reports and dashboards in a company workspace, (3) automate dataflows and schedule updates, (4) share and collaborate with peers and teams.

> **IMPORTANT**
>
> The data is adapted from the Los Angeles County Environmental Health site and is for informational purposes only. The inspectors and email addresses are fictitious (deck p.6).

The data model (7 tables)

Source: Deck p.7

Table	Type	What it holds
Restaurant Inspection Data	Fact	Inspection and violation data: date, facility, geo data, grade, etc.
Calendar	Date	Date, month, week, quarter and year details.
City	Lookup	Two columns: facility city and region id.
Inspector	Lookup	Multi-column: inspector name, email, assigned region, role, etc.
Region	Lookup	Two columns: facility region and associated region id.
Restaurant Name	Lookup	One column: unique facility name.
Sanitary Grade	Lookup	One column: A, B or C.



The six lookup tables each connect to the fact table. The exact column pairs are listed in Part 8.

The people in the story

Person	Role	Asks you to...
David Schmitt	Lead Inspector	Create the workspace and load the CSV (Parts 4 and 5); later, widen his data access (Part 11).
Susie Salt	Inspector	Build out the dataflow (Part 7) and finish the semantic model (Part 8).
Dillan Krueger	Inspector	Build the Restaurant Inspections report (Part 9).
Samantha Cook	Chief Inspector	Assemble the dashboard (Part 9).
Brent Finnegan	Inspector	Set up alerts (Part 9) and export to PowerPoint (Part 10).

Part 2. Concepts you need first

Microsoft Fabric and where Power BI fits

Source: Deck p.10-14

Microsoft Fabric is an end-to-end data and analytics solution that brings together new and existing components from Power BI, Azure Synapse and Azure Data Factory in one integrated environment. All data lives in **OneLake**, on a SaaS foundation.

Fabric component	What it does
Power BI	Visualize, analyze and share data in a centralized online environment.
Data Factory	ETL with Dataflows and Data pipelines; create, schedule and coordinate data transformations.
Data Activator	Business users define rules and conditions that trigger data-driven alerts and automated actions.
Industry Solutions	Industry-specific connectors, transformations and tools.
Data Engineering	Lakehouses, Notebooks and Spark Jobs to build and maintain data infrastructure.
Data Science	Notebooks, Experiments and ML models to prep data and build predictive insights.
Data Warehouse	Unified platform for schemas, tables, stored procedures and other database objects.
Real-Time Analytics	Ingest, transform and query any data format, from 1 GB to 1 PB, then visualize and share.

> **PRO TIP**
>
> Fabric is huge. Build expertise in the few areas that fit your role instead of trying to master everything (deck p.11).

Power BI Desktop vs Power BI Service

Source: Deck p.15-16

Power BI Service is a cloud-based BI service for visualizing, analyzing and sharing data online. It can create interactive reports and dashboards, connect to online and on-premises sources, support collaboration in shared workspaces, scale across an organization and integrate with other Microsoft products. The two tools have become more similar over time.

Stage	Power BI Desktop	Power BI Service
Connect	190+ data sources, Power Query, M code	160+ data sources, Dataflows (Power Query), M code, Gateways
Create	Data models, relationships, DAX calculations, row-level security	Data models, relationships, DAX calculations, row-level security
Build	Reports and dashboards, visualization tools, mobile layout	Reports and dashboards, visualization tools, mobile layout
Share	Publish to Service, share file directly	Workspaces, Apps, Publish to Web, Export options

Licences and capacities

Source: Deck p.17-19

Licence	What you get
Power BI Free	No licence required but limited. Connect to 190+ sources, publish to web, export to PowerPoint/Excel/CSV, share to Pro or Premium users. Cannot collaborate or distribute content.
Power BI Pro	Everything in Free plus user-based licence, workspaces, email subscriptions, peer-to-peer sharing, publish apps, Analyze in Excel and Desktop, paginated reports.
Premium Per User (PPU)	Everything in Pro plus AI capabilities, advanced dataflows, deployment pipeline, datamarts, XMLA endpoint connectivity, enhanced automatic page refresh, increased storage limits.

Limit	Pro	PPU	Premium
Model memory size limit	1 GB	100 GB	400 GB
Semantic model refreshes	8 per day	48 per day	48 per day
Maximum storage (Power BI native)	10 GB / user	100 TB	100 TB

Capacity tiers for organizations are Power BI Premium, Fabric Reservation and Fabric Pay-As-You-Go. They share these traits: capacity-based licences (P/F SKUs), people can consume company reports without individual licences, a Pro or PPU licence is still required to create, publish, share and collaborate in a workspace, and they give greater scale and unified access to Fabric workloads (deck p.18).

> **IMPORTANT**
>
> Viewing rights: when you share a report or dashboard or publish an app, each viewer needs a paid licence (Pro or PPU) or access to a premium capacity (Premium or Fabric). Pro users can share with PPU users, but not the other way around (deck p.102).

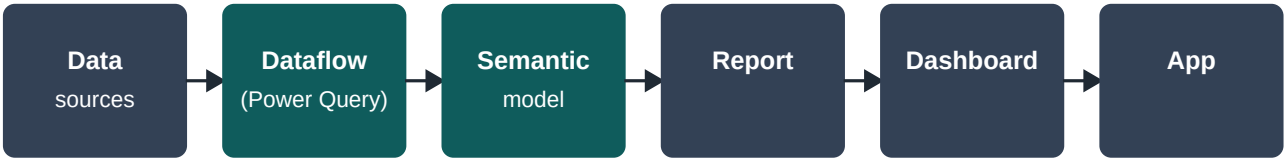
The three Power BI personas

Source: Deck p.20

Persona	What they do
Creator	Designs, builds and maintains reports and dashboards, from data wrangling to modeling and visualization. Needs a deep, well-rounded skillset.
Analyzer	Explores and manipulates data to find insights; edits visuals, writes DAX measures and columns, uses Q&A, interactions, report filters and bookmarks.
Collaborator	Publishes and shares reports, apps and workspaces; knows user roles, row-level security and app/workspace setup.

The Power BI Service workflow

Source: Deck p.29, 67



Data lineage: how data flows from its source to the people who use it (the app is built from reports and dashboards).

Inside one workspace the flow is: **Dataflows > Semantic model > Reports > Dashboards**. Your project follows exactly this order.

Vocabulary

Term	Meaning
Workspace	Container for dataflows, semantic models, paginated reports, reports and dashboards where you collaborate.
Dataflow	Cloud-based data prep (Power Query in the cloud).
Semantic model	Structured representation of data that defines the relationships, calculations and formats needed to build reports (formerly called a dataset).
Report	A collection of visuals built on a semantic model.
Dashboard	A single page that aggregates visuals from different sources/reports.
Paginated report	A report designed for printing or PDF generation.
App	A packaged set of reports and dashboards for a large audience.

Part 3. Account and trial setup

Step-by-step: create your Power BI account

Source: Deck p.21

1. Open **powerbi.microsoft.com/landing/free-account** and click **TRY FOR FREE**. (If you already have an account, skip to step 3.)
2. Follow the account set-up process using a **work or school email address**.
3. When the account exists, go directly to **app.powerbi.com** and sign in.

> **IMPORTANT**
>
> Power BI requires a work or school email address. Domains like @gmail, @yahoo and @outlook will not be accepted.

No work or school email? Use Maven Analytics

Source: Deck p.22

1. Create a free Maven Analytics account (or log in) at **https://www.mavenanalytics.io**.
2. Open the **External Apps** menu and activate the **Power BI Service** trial; follow the steps.
3. Sign in at **app.powerbi.com** with the new **@mavenbi.org** email account.

> **IMPORTANT**
>
> Microsoft keeps changing trial options, limits and settings. You may or may not get Fabric items as part of the free trial.

Activate the Fabric trial

Source: Deck p.23

1. Sign in to Power BI Service.
2. Click your **profile picture (top-right corner)**. The panel shows your name, email and **License type: Free account**.
3. Click **Start trial**.
4. In the dialog **Upgrade to a free Microsoft Fabric trial**, read the note and click **Start trial** again (the other button is Cancel).

CHECK

The header bar now shows **Trial: 59 days left** (the trial lasts 60 days; the counter appears every time you log in). Expected result: a trial capacity you can assign to a workspace.

IMPORTANT

A Fabric trial includes tools that this course does not cover. Check Microsoft's site for post-trial pricing (powerbi.microsoft.com).

Part 4. The interface and your first workspace

Tour of the Power BI Service interface

Source: Deck p.24

Area	What it does
Office 365 app launcher (top-left)	Email, OneDrive, Calendar, etc.
Navigation path (breadcrumb)	Highlights the active Fabric component, workspace, app, report or dashboard.
Navigation pane (left)	Home, Create, Browse, OneLake data hub, Apps, Metrics, Workspaces, My workspace.
Fabric components (bottom-left)	Access and switch between Fabric components such as Power BI.
Workspace actions (toolbar)	Get data, add new items, update settings, manage access.
Settings menu (top-right gear)	Quick links to settings, help, feedback, downloads.
Content canvas (centre)	Shows all your semantic models, visuals, reports, dashboards and workbooks.

My workspace vs a shared workspace

Source: Deck p.25

	My workspace	Workspace
Who	Personal, one user (free or paid licence)	Shared by many users (paid licences or Premium capacity)
Collaboration	Only you can access; cannot collaborate	Many users access and collaborate on content
Sharing	Content can be shared with individual Pro/PPU users (shows under 'Shared with me')	Content created and shared across the organization
Contains	Semantic models, reports, dashboards	Semantic models, reports, dashboards plus dataflows

IMPORTANT

Build the project in a shared **Workspace**, not My workspace. Dataflows, scheduled refresh for teams, apps and role-based sharing all need it.

Step-by-step: create the company workspace

Source: Deck p.26, 35

1. In the **navigation pane**, click **Workspaces**.
2. At the bottom of the flyout, click **+ New workspace**.
3. In **Create a workspace**, type a **Name** (required). A green note says "This name is available". Example in the deck: *My Example Workspace*; the project workspace shows as **Restaurant Inspections**.
4. Optional: add a **Description** (deck example: "This is an example workspace for LA restaurant inspectors"), a **Domain**, and a **Workspace image** (Upload).
5. Click **Advanced** to open the licence options. Choose a **License mode**: Pro, Premium per-user, Premium capacity, Fabric capacity, Embedded or **Trial**. For this course use **Trial**. Here you can also choose the model storage format (**small** or **large**).
6. Click **Apply**.

> **CHECK**
>
> The workspace appears in the Workspaces list with a diamond icon next to its name, and you land inside it.

Workspace settings

Source: Deck p.27

Open **Workspace settings** from the workspace you just created. The left menu lists: About, Premium, Azure connections, System storage, Git integration, OneLake, Workspace identity, Network security, Other; and under the Power BI heading: General, Data connections, Embed codes.

- **Premium** tab: change the **license mode** (Pro, Premium per-user, Premium capacity, Fabric capacity, Trial, etc.) and the semantic model storage format.
- **Power BI > General** tab: scroll to **Data model settings** and tick **Users can edit data models in the Power BI service (preview)**. This lets workspace members edit semantic models directly in Service (define relationships, DAX, roles). Other options on this tab include Secure update and Template apps.

> **EXTRA**
>
> The deck does not show which button opens Workspace settings. In the current interface it is in the workspace header (a gear or the ... menu). If you cannot find it, use the search box at the top.

Workspace actions you will use all project

Source: Deck p.28

Item / action	Meaning
Reports	Collection of visuals built on a semantic model.
Dashboards	Single-page display of visuals from different sources.
Paginated report	Built for printing or PDF.
Semantic models	Defines relationships, calculations and formats.
Dataflows	Cloud-based data prep (Power Query in the cloud).
Share	Share content with colleagues.
Favorite	Add selected content to Favorites.
Refresh now / Scheduled refresh	Refresh source data now / set an automated refresh schedule.
More options (...)	Contextual settings for the selected item.

Workspace folders (organizing content)

Source: Deck p.60

1. In the workspace, click **+ New > Folder (Preview)** and name it (the deck example folder is *Fuzzy Factory*).
2. Select the content types you want inside it and move them (use the item's **... > Move to**).
3. Open the folder to see similar content together; the breadcrumb reads **Workspace > folder name**.

Part 5. Connecting to data and loading your first file

Ways to bring data into Service

Source: Deck p.32

Route	Options
Discover existing content	**OneLake data hub** (built on Azure Data Lake Storage Gen 2; stores Fabric data like lakehouses, warehouses and datamarts in Delta-Parquet format) and **Apps** (collections of reports and dashboards other people in your organization created).
Create new content: New	**+ New > Semantic model** to import CSV or Excel files or paste/enter data manually; connect Azure databases; build a **Dataflow**.
Create new content: Upload	Upload PBIX, CSV or XLSX from OneDrive, SharePoint or your local file system (OneDrive and SharePoint need an organization account).

Step-by-step: load Restaurant Inspection Data.csv

Source: Deck p.33, 35

1. Open your workspace.
2. Click **+ New**, then choose **Semantic model**.
3. On **Add data to start building a report**, choose **CSV** (the other tiles are **Excel** and **Paste or manually enter data**).
4. Select **Restaurant Inspection Data.csv** and follow the prompts to load it.
5. Back in the workspace, a new item named **Restaurant Inspection Data** appears with type **Semantic model**. Click it to open the **Semantic model view**.

> **IMPORTANT**
>
> Data imported from Excel must be formatted as a table.

The semantic model view

Source: Deck p.34

Area	What it does
Menu options (top bar)	File, Refresh, Share, Explore this data, Export, Lineage, Open data model: manage permissions and settings, refresh, share, explore, export.
Model details	Title, description (**+ Add description**), location (workspace) and refresh date.
Explore data tile	**Explore this data** builds visuals, auto-creates reports or creates blank reports. The drop-down arrow gives more choices.
Share data tile	**Share semantic model** grants access and sets permissions.
Tables pane (right)	Tick tables and columns to view and export the data. Selecting more than one table lets you **Create paginated report**.
Related items	Items using the same source (reports, dataflows, dashboards) appear here.

Assignment 1: load the CSV and profile the data

Source: Deck p.35-36

1. Create a new workspace with a name, a brief description and the right licence mode (Part 4).
2. Load **Restaurant Inspection Data.csv** into the workspace (steps above).
3. Open the semantic model and use **Explore this data** or the Tables pane to profile the columns, then answer the three questions below.

Question	Answer (deck solution)
What are the 3 types of grades?	A, B, C
What does the violation_status field contain?	Two text values: "Out of Compliance" and "Violation"
How many facilities are in the dataset?	44,031 (a distinct count)

> **EXTRA**
>
> To reproduce the 44,031, put a distinct count of the facility identifier column (the dataset has facility_id and facility_name columns, deck p.51) into a visual in Explore this data. If your count differs, check you are counting distinct values and not rows.

Part 6. Data gateways

What a gateway is

Source: Deck p.37-38

A **data gateway** is a bridge that connects on-premises data sources to Power BI without moving the data to the cloud. It keeps data on your network (no duplication), enables scheduled refresh, secures the data transfer and works with other cloud services (Analysis Services, Data Factory, etc.).

- Gateways are currently supported only on **Windows** machines.
- To use scheduled refresh with Excel, the file must contain a **data model**.

Capability	Personal mode	Standard mode
Compatible cloud services	Power BI only	Power BI, Power Apps, Power Automate, Azure Logic Apps, Azure Analysis Services, Dataflows
Multi-user support	No	Yes
Runs as an app for non-administrators	Yes	No
Runs as a single user	Yes	No
Import data and set up scheduled refresh	Yes	Yes
DirectQuery support	No	Yes
Live connection to Analysis Services	No	Yes

PRO TIP

Use the **Standard mode** gateway. It does everything personal mode does, supports multiple cloud services, and can be shared with others in your organization.

Step-by-step: install a standard gateway

Source: Deck p.39

1. In Power BI Service, open the **Download** menu (arrow icon, top-right) and choose **Data Gateway**, or go to **powerbi.microsoft.com/gateway** and click **Download Standard Mode**.
2. Run the installer and follow the installation instructions.
3. Sign in with your **Power BI Service credentials**.
4. Create a **gateway name** and a **recovery key** and finish. Store the recovery key safely.

> **EXTRA**
>
> For system requirements see https://docs.microsoft.com/en-us/data-integration/gateway/service-gateway-install (linked on deck p.38). The PC running the gateway must be on whenever a refresh is due.

Step-by-step: confirm the gateway is online

Source: Deck p.40

1. Click the **gear icon** (top-right) in Power BI Service.
2. Choose **Manage connections and gateways**.
3. Under **On-premises data gateways** find your gateway; its status should read **Online**.

> **PRO TIP**
>
> If the status shows a warning, click the status icon to see troubleshooting tips.

Part 7. Dataflows

What dataflows are and why use them

Source: Deck p.41-43

Dataflows are cloud-based data pipelines (Power Query in the cloud) used to extract, transform, store and integrate data from many sources.

- Reuse queries across semantic models and other dataflows.
- Run ETL (extract, transform, load) on specific schedules.
- Create a centralized store as a single source of truth.
- Create, manage and share reusable transformation logic.

Implementation in three moves: (1) create and build the dataflow in Service, (2) refresh it, (3) connect to it as the data source in Power BI Desktop and publish as a semantic model.

Use case	Example from the deck
1. Reusable logic across many semantic models	One dataflow instead of re-creating the connection and applied steps in every report.
2. Schedules that match reporting cadence	One dataflow refreshing at 8 am for an internal KPI report, another weekly for an executive client dashboard.
3. Single source of truth and centralized ETL	Create, manage and QA ETL processes in one central place instead of separate error-prone ones.

Step 1. Create the dataflow and pick a source

Source: Deck p.44

1. In the workspace click **+ New > Dataflow**.
2. On **Start creating your dataflow** you see four tiles: **Define new tables** (Add new tables), **Link tables from other dataflows** (Add linked tables), **Import Model**, **Attach a Common Data Model folder (preview)** (Create and attach). Choose **Define new tables > Add new tables**.
3. On **Choose data source** pick **Text/CSV** (other options shown include Excel workbook, SQL Server database, Web API, Dataflows, Dataverse, Azure SQL database, Power BI dataflows (Legacy)).

> **IMPORTANT**
>
> This process is for **Dataflows Gen1**. Dataflows Gen2 are a Fabric item and may not be available to everyone (deck p.44).

Step 2. Enter connection settings and credentials

Source: Deck p.45-47

1. Under **Connection settings** choose **Link to file** or **Upload file (Preview)**. With Link to file, type or paste the complete file path **without quotes or tick marks** (or click **Browse OneDrive...**). With Upload file, drag the file in or click **Browse...**.
2. **Connection:** leave **Create new connection**. The **Connection name** fills in from the path.
3. **Data gateway:** select an installed, running gateway (deck example: *[On-premises] Maven Analytics BI*).
4. **Authentication kind:** choose the type. Options shown: Anonymous, Basic, Organizational account, Service principal; with a gateway and a local file the deck uses **Windows**.
5. Enter **Username** (format domain\alias) and **Password**. **Privacy level** can stay at None.
6. Click **Next**.

> **IMPORTANT**
>
> Access to OneDrive for Business is required to upload files or to browse OneDrive (deck p.45).

How to find the Windows username and password to enter

- **Option 1:** enter a Microsoft administrator account username and password that can sign you in to all apps. You may need to adjust password settings under **Account > Sign-in options** (Windows 11). Path: **Windows Settings > Accounts > Email & Accounts** (Accounts used by other apps).
- **Option 2:** open **Command Prompt**, type `whoami` and press Enter. The result looks like `computername\username` (deck example: aaronpc\ampar). Use that as the username.

> **IMPORTANT**
>
> If you sign in to your computer with a PIN or biometrics (fingerprint or face), you may need to use your Microsoft account password here instead (deck p.47).

Step 3. Clean and shape the data in Power Query

Source: Deck p.48

1. After Next, the **Preview file data** page opens. Click **Transform data** (bottom-right).
2. The cloud **Power Query Editor** opens. Applied steps in the deck example are **Source**, **Promoted headers** and **Changed column type**.
3. Clean, transform and shape the data. In **Query settings > Properties** set the table **Name** (deck: *Restaurant Inspection Data*).

Step 4. Name, save and refresh

Source: Deck p.49

1. Click **Save & close** in Power Query to validate and process the query.
2. In **Save your dataflow**, enter a **Name** (deck example: *Inspection Services*) and an optional **Description** (deck: "Dataset contains restaurant specific records, employee data, violation codes and status, owner data, and more"). Click **Save**.
3. A **Success** message says the dataflow has been saved. Click **Refresh now** to "lock in" the dataflow (the other button is **Set a refresh schedule**).

> **IMPORTANT**
>
> You MUST refresh the dataflow before it can be used as a data source in Power BI.

The dataflow interface

Source: Deck p.50

Control	What it does
Edit tables	Opens Power Query in the cloud to edit/update.
Add tables	Adds a new or linked table to the dataflow.
Table name (expand)	Shows the list of columns in the table.
Actions: Edit table	Opens Power Query in the cloud linked to the selected entity.
Actions: Properties	Add a description for the table to give context.
Actions: Incremental refresh	Configure an incremental refresh policy (row storage and refresh settings). Requires a Premium workspace.

Assignment 2: build out the Inspection Team dataflow

Source: Deck p.52-54

> **EXTRA**
>
> The deck's solution shows these seven tables in the dataflow: Restaurant Inspection Data, Calendar Lookup, City Lookup, Inspector Lookup, Region Lookup, Restaurant Name Lookup, Sanitary Grade Lookup.

1. **Add the remaining six tables** to the Inspection Team dataflow: calendar, city, inspector, region, restaurant name and sanitary grade. Use **Edit tables** or **Add tables** and the same Text/CSV, gateway and credential steps.
2. **Save, close and refresh** the dataflow.
3. **Edit the dataflow** again and add these transformations to the **Calendar** table:
   - Set **Transaction_Date** as a **Date** data type.
   - Add calculated columns: **Year**, **Quarter of Year**, **Month**, **Start of Year**, **Start of Month**, **Start of Week**.
4. In the **Restaurant Name** and **Sanitary Grade** tables:
   - Apply **Promote first rows to header** (Power Query: Home > Use first row as headers).
5. **Save & close**, then **refresh the dataflow from the Workspace view** (use the refresh icon on the dataflow's row).
6. **Connect from Power BI Desktop** using the Restaurant Inspections workbook:
   - Disable **automatically create relationships** before loading (so Desktop does not guess relationships).
   - **Get data > Dataflows**, connect to the **Inspection Team** dataflow and import the **6 tables** from step 1.
   - Save your work.

> **EXTRA**
>
> In Power Query the date columns are under the **Add column** tab: select Transaction_Date, then **Date** and pick Year, Quarter > Quarter of Year, Month > Month, Start of Year, Start of Month and Start of Week. To switch off auto-relationships in Desktop: **File > Options and settings > Options > Current File > Data Load**, untick the relationship auto-detect option (wording varies by version).

Connect to the dataflow in Power BI Desktop

Source: Deck p.51

1. In Desktop open the **Home** tab and click **Get data**.
2. Choose **Dataflows** (it sits in the Common data sources list). Sign in if asked.
3. In the **Navigator**, expand **Dataflows > Workspaces > your workspace > Inspection Services** and tick the tables you want.
4. Click **Load** (or **Transform Data** to keep editing).

> **CHECK**
>
> The Data pane in Desktop lists the loaded table(s) and their columns, e.g. activity_date, employee_id, facility_address, facility_city, facility_id, facility_name, facility_state, facility_zip, grade, owner_id, owner_name and more.

Other dataflow sources and Gen1 vs Gen2

Source: Deck p.55-57

- **Cloud databases** (Azure, AWS, Google Cloud) can be used as dataflow sources (p.55).
- **Local MySQL** (e.g. MySQL Community Server) works through a gateway on your computer: supply the local host address and database name, pick the running gateway and enter the server username and password. You do not need to pay for a cloud database just to use MySQL (p.56). Setting up Community Server is NOT part of the course.
- **Dataflow Gen2** (the new generation, reached from Data Factory) coexists with Gen1. Per the deck it adds: a shorter authoring flow, auto-save and background publishing, data destinations (Azure, Lakehouse, Warehouse), improved monitoring and refresh history, data pipelines integration and high-scale compute. **Both** generations let you author with Power Query and Get Data via the dataflows connector.
- The deck marks three features as **Gen1 only**: Direct Query via the dataflow connector, incremental refresh and AI Insights support. This guide uses Gen1. Details: learn.microsoft.com/en-us/fabric/data-factory/dataflows-gen2-overview (p.57).

Part 8. Scheduled refresh and the semantic model

Schedule a dataflow refresh

Source: Deck p.58

1. Click the **gear icon > Power BI settings** (under Resources and extensions), or use the **Schedule refresh** shortcut icon on the dataflow's row in the workspace.
2. Open the **Dataflows** tab and expand **Refresh**.
3. Switch **Configure a refresh schedule** to **On**.
4. Choose the **Refresh frequency**: **Daily** or **Weekly**, and the **Time zone**.
5. Under **Time** set the hour, minute and AM/PM. Click **Add another time** for more runs. Pro can add up to **8 refreshes per day**, PPU up to **48**.
6. Under **Send refresh failure notifications to** tick **Dataflow owner** and/or **These contacts** (enter email addresses).
7. Click **Apply**.

Schedule a semantic model refresh

Source: Deck p.59

1. In **Power BI settings** open the **Semantic models** tab and select your semantic model.
2. Expand **Refresh** and switch the schedule **On**.
3. Set **Frequency** (Daily or Weekly), **Time zone** and **Time** (half-hour increments). Use **Add another time** for extra runs.
4. Tick **Semantic model owner** and/or **These contacts** for failure notifications, then click **Apply**.

> **PRO TIP**
>
> If you use both schedules, **offset** them. The deck example runs the dataflow at 8:00 AM and the semantic model at 8:30 AM. If they run at the same time the report may not show the latest updates.

> **CHECK**
>
> In the workspace list the **Refreshed** and **Next refresh** columns show when the dataflow and semantic model last ran and will run next. Confirm the two schedules align and there are no errors.

A real-world automation setup

Source: Deck p.61

Layer	Example
Relational database service	MySQL on Amazon RDS (database installed and reachable in the cloud).
Always-on virtual computer	Amazon EC2 running the data gateway (a computer that is always on).
Power BI Service	Dataflows and semantic models with scheduled refresh configured.

Assignment 3: finish the semantic model in Desktop

Source: Deck p.62-63

Step A. Create the relationships

1. In Desktop open **Model view**.
2. Connect **Restaurant Inspection Data** to every lookup table by dragging the lookup column onto the matching column in the fact table (or use **Manage relationships > New**). The deck's finished relationships are below.
3. Confirm every relationship is **one-to-many** (shown from the fact table as many-to-one), with a **single** cross-filter direction flowing downstream from the lookup (one side) to the data table (many side).

From: Table (column)	To: Table (column)
Restaurant Inspection Data (activity_date)	Calendar Lookup (Transaction_Date)
Restaurant Inspection Data (facility_city)	City Lookup (facility_city)
Restaurant Inspection Data (facility_name)	Restaurant Name Lookup (facility_name)
Restaurant Inspection Data (grade)	Sanitary Grade Lookup (sanitation_grade)
Restaurant Inspection Data (region_id)	Inspector Lookup (assigned_region)
Restaurant Inspection Data (region_id)	Region Lookup (region_id)

> **CHECK**
>
> Open Manage relationships: all six rows show **Active** ticked.

Step B. Set facility_zip as a category

1 Open **Table view** and click the **facility_zip** column in Restaurant Inspection Data.

2 On the **Column tools** ribbon set **Data category** to **Postal code**.

> **EXTRA**
>
> The deck screenshot shows Summarization = Sum on this column. If zip codes ever get summed in a visual, set Summarization to **Don't summarize**.

Step C. Create the DAX measures

Create the measures in a table called **Measures Table** (the name shown in the deck's Data pane). The deck lists these measures:

Measure	Purpose
[Total Inspections] and [All Inspections]	Inspections in the current filter vs. all inspections ignoring filters.
[% of All Inspections]	Share of all inspections.
[Prev. Month Total Inspections]	Inspections one month earlier (used as the KPI goal).
[Total Violations] and [All Violations]	Violations in the current filter vs. all.
[% of All Violations]	Share of all violations (shown on tooltips).
[Average Sanitary Grade] and [Prev. Month Avg Sanitary Grade]	Average score now and one month earlier.
[Grade A Inspections], [Grade B Inspections], [Grade C Inspections]	Inspections by grade.

> **VERIFY**
>
> The deck does not print these formulas. The ones below are suggested starting points; the column names total/score/grade must match your table. If your table has an inspection-ID column, use DISTINCTCOUNT of it for [Total Inspections] instead of COUNTROWS.

```
Total Inspections = COUNTROWS('Restaurant Inspection Data')
All Inspections = CALCULATE([Total Inspections], ALL('Restaurant Inspection Data'))
% of All Inspections = DIVIDE([Total Inspections], [All Inspections])
Prev. Month Total Inspections =
    CALCULATE([Total Inspections], DATEADD('Calendar Lookup'[Transaction_Date], -1, MONT

Total Violations = COUNTROWS('Restaurant Inspection Data')   -- if each row is one viola
All Violations = CALCULATE([Total Violations], ALL('Restaurant Inspection Data'))
% of All Violations = DIVIDE([Total Violations], [All Violations])

Average Sanitary Grade = AVERAGE('Restaurant Inspection Data'[score])
Prev. Month Avg Sanitary Grade =
    CALCULATE([Average Sanitary Grade], DATEADD('Calendar Lookup'[Transaction_Date], -1,

Grade A Inspections = CALCULATE([Total Inspections], 'Restaurant Inspection Data'[grade]
Grade B Inspections = CALCULATE([Total Inspections], 'Restaurant Inspection Data'[grade]
Grade C Inspections = CALCULATE([Total Inspections], 'Restaurant Inspection Data'[grade]
```

EXTRA

To create the Measures Table: Home > Enter data, name it Measures Table and click Load. Then select it and use Home > New measure for each formula. Mark the Calendar Lookup table as a date table (Table tools > Mark as date table, using Transaction_Date) so the previous-month measures work.

Step D. Save and publish

1. **Save** the Desktop file.
2. On the **Home** ribbon click **Publish**.
3. Choose your workspace (for example **Restaurant Inspections**) and click **Select**.
4. When it finishes, open the workspace: the **semantic model** and **report** now appear there.

Part 9. Reports, dashboards and insights

Data lineage view

Source: Deck p.67-69

Data lineage is the flow of data from its source to destinations like dashboards or apps: data sources > dataflows > semantic models > reports > dashboards > apps. The workspace can be shown as a **List view** or a **Lineage view** (two small icons at the top-right of the workspace list, beside the Filter button).

Lineage tool	What it does
Impact (icon on a data source)	Shows the impact of the data source across the workspace (child items, workspaces).
Refresh icon	Refreshes the dataflow or semantic model for that card.
Lineage path icon	Highlights the data lineage path based on your selection.
Item menu	Opens a menu specific to the selected lineage item (e.g. Semantic model details > View item details).

PRO TIP

Use the lineage path tool to diagnose errors in reports and dashboards that get data from several dataflows, especially across workspaces.

Reading view vs Editing view

Source: Deck p.70-73

	Reading view	Editing view
Use for	Exploring, filtering and pinning tiles and pages to dashboards	Editing, modifying and creating report visuals
Left/bottom	Pages pane to navigate pages	Report pages along the bottom (each a blank canvas, as in Desktop)
Right side	Filters pane (visual, page, report level and drill-through)	Visualizations and Format pane (charts, slicers, maps, matrices) plus Data pane

Reading view toolbar

- **File:** Save a copy, Download this file, Manage permissions, Print this page, Embed report, Generate a QR code, Settings.
- **Export:** Analyze in Excel, PowerPoint, PDF.

- **Share**, **Chat in Teams** (launch and share to Microsoft Teams), **Explore this data** (new data exploration window), **Get insights**, **Subscribe to report**, **Edit** (launch editing mode).
- **... menu:** See related content, Open lineage view, Open usage metrics, Pin to a dashboard, View semantic model.
- Small icons: **Reset to author's default view**, **Personal bookmarks**, **View/contrast/visual type**, and **Refresh, Comment, Favorite**.

Editing view toolbar

- **File:** Save, Save as, Print, Embed, Embed in SharePoint Online, Publish to web, Export to PowerPoint, Export to PDF, Download this file. **View:** Fit to page/width, Actual size, High-contrast colors, Smart guides, Gridlines, Snap to grid, Lock objects, Selection pane, Bookmarks pane, Sync slicers pane, Copilot pane, Insights pane.
- **Reading view**, **Mobile layout**, **Open data model** (edit relationships, DAX measures, roles in Service; the feature must be enabled in your tenant).
- **Explore this data**, **Ask a question**, **Data/drill** (show as a table, show next level, expand, drill down/through), **Text box**, **Shapes**, **Buttons** (arrows, reset, back, info, help, Q&A, bookmark, blank, apply/clear all slicers, navigator), **Visual interactions** (Edit interactions, Drilling filters other visuals), **Refresh**, **Save**, **Pin to a dashboard**, **Chat in Teams**, **...** (Generate a QR code, Analyze in Excel, Get insights).

Building and formatting visuals in Service

Source: Deck p.74-76

- **Build** menu: change visual type, import new visuals and add fields to chart components (x-axis, y-axis, legend, tooltips, etc.). It is contextual: you only see options relevant to the selected visual.
- **Format** menu: add common elements (title, axis labels, data labels, legend) and reach the Format pane for more.
- **Analytics** menu: reference lines, forecasting, trend lines, statistical summaries.
- Two ways to build a visual: insert a chart type and add data, or drag a field from the Data pane onto the canvas.

> **IMPORTANT**
>
> Options and menu settings differ between Desktop and Service.

Adding an image (logo) in Service

1. Select any visual with a plot area (bar, line, area...) and add a dimension to the axis.
2. In the **Format pane**, add an image from file to the plot area and set **Image Fit** to **Fill**.
3. Remove titles and labels, format the background and resize.

> **PRO TIP**
>
> If you plan to use images, add them in Power BI Desktop first. It is much simpler there.

Assignment 4: build the Restaurant Inspections report

Source: Deck p.77-79

Start from the published semantic model: in the **semantic model view** open the **Explore this data** drop-down and choose to create a blank report, or use **+ New > Report** in the workspace and pick the semantic model. Then switch to **Edit**.

Page 1: Inspections & Violations

Element	How to build it
Logo	Maven logo via the image technique above (or add it in Desktop).
Slicer: date	Transaction_Date from Calendar Lookup (between/range slider; the deck range is 7/1/2015 to 12/31/2017).
Slicer: inspector	Inspector name from Inspector Lookup (drop-down).
KPI card: Current Month Inspections	Value = [Total Inspections]; goal = [Prev. Month Total Inspections]. The deck shows 5,901 with goal 6,061 (-2.64%).
KPI card: Avg. Sanitation Grade	Value = [Average Sanitary Grade]; goal = [Prev. Month Avg Sanitary Grade]. The deck shows 92.21 with goal 91.91 (+0.33%).
Bar chart: Inspection Volume by County & Restaurant Name	Axis: facility_region (county) then facility_name, so you can **drill down to restaurant name**; Values: [Total Inspections].
Decomposition tree: Violations by Region & Type	Analyze: [Total Violations] (Number of Violations). Explain by: facility_region (county), then the violation description. Put [% of All Violations] on the **tooltips**.
Column chart: Violations Trending	Axis: month from Calendar Lookup; Values: [Total Violations].
Map: Restaurant Zip Code	Location: facility_zip; Size: [Total Violations].
Donut chart: Sanitation Grade	Legend: grade; Values: [Total Inspections].

Page 2: Zip Code Detail

Element	How to build it
Slicer	facility_region (list style).
Azure Map	**Location** = facility zip (with state); **Size** = Number of Violations.
Column chart	Violations by month.

1. Rename page 1 to **Inspections & Violations**. Add a new page with the **+** at the bottom and name it **Zip Code Detail**.
2. Add the visuals from the tables above. For each one use the **Build** menu to place fields, then **Format** for title and labels.
3. Adjust formatting and layout as you see fit (the deck uses black title bars with white text, teal bars and dark-red columns).
4. Click **Save** and give the report a name (**Restaurant Inspections**).

> **IMPORTANT**
>
> Slicers cannot be pinned individually to a dashboard, but they come along when you pin a whole live page (see below).

Personal bookmarks

Source: Deck p.80

Personal bookmarks capture a static state of a visual or report page. They work like Desktop bookmarks but are visible **only to you**. Create them from Reading or Editing view with the **Bookmark** icon in the top bar.

Pin tiles and live pages to a dashboard

Source: Deck p.81-82, 86-88

Pin a single visual (tile)

1. In the report hover over the visual and click the **pin icon**.
2. In **Pin to dashboard** choose **Existing dashboard** or **New dashboard**. For a new one type a **Dashboard name** (deck: *Restaurant Inspections Dash*).
3. Click **Pin**.

Pin a live page

1. In **Reading view** click **... > Pin to a dashboard**.
2. Choose **Existing dashboard** (or New) and click **Pin live**.

> **PRO TIP**
>
> Pinning a live page imports slicers, preserves functionality and reflects any later updates to the report.

Assignment 5: assemble the dashboard

1. Pin the **Zip Code Detail** page as a **live page** to the **Restaurant Inspections** dashboard.
2. Pin these tiles to the same dashboard:
   - Current Month Inspections KPI card
   - Avg. Sanitation Grade KPI card
   - Sanitation Grade donut chart
   - Inspection Volume by County bar chart
3. Open the dashboard and **arrange and resize** the tiles as you see fit (drag to move, drag the corner to resize).

The dashboard interface and tools

Source: Deck p.83-85

- **Ask a question about your data:** natural-language box that generates new tiles.
- **File:** Save a copy, Manage permissions, Print this page, Settings.
- **Share**, **Chat in Teams**, **Comment** (comment on specific visuals), **Subscribe to dashboard**.
- **Edit:** Add a tile, Dashboard theme, Mobile view.
- **... menu:** See related content, Open lineage view, Open usage metrics.
- **Share dashboard** dialog: enter names/emails; tick **Allow recipients to share this dashboard**, **Allow recipients to build content with the data associated with this dashboard**, **Send an email notification**; add a message; click **Grant access**.

Embed web content as a tile

Use **Edit > Add a tile** and choose web content, then paste this code with your own address in `src`:

```
<iframe width="100%" height="100%"
src=https://www.mavenanalytics.io
frameborder="0" allowfullscreen></iframe>
```

PRO TIP

You can tweak the HTML to customize the style of embedded content (see Microsoft Docs for options).

Data-driven alerts

Source: Deck p.89-91

1. On the dashboard, click the **ellipsis (...)** next to the tile and choose **Manage alerts**.
2. Click **+ Add alert rule**.
3. Switch **Active** to **On** and type an **Alert title**.
4. Choose the **Condition** (Above or Below) and type the **Threshold**.
5. Choose **Maximum notification frequency**: **At most every 24 hours** or **At most once an hour**.
6. Leave **Send me email, too** ticked if you want email as well as notification-centre messages.
7. Click **Save and close**.

Alert	Tile	Title	Condition	Threshold	Frequency
1	Average Sanitation Grade KPI card (measure: Average Sanitary Grade)	Average Sanitation Grade below 92	Below	92	At most every 24 hours
2	Current Month Inspections KPI card (measure: Number of Inspections)	Exceeded 6500 Inspections	Above	6500	At most every 24 hours

IMPORTANT

Alerts can ONLY be configured on gauge charts, KPI cards or cards. They do not work on live pinned report pages. Alerts are only sent if your data changes.

Natural language Q&A

Source: Deck p.92

1. On the dashboard click **Ask a question about your data**.
2. Type a question such as `top 10 violations by restaurant`. Power BI builds a bar chart (here: the 10 restaurants with the most violations).
3. Click **Pin visual** to add the Q&A result to the dashboard.

> **PRO TIP**
>
> Improve Q&A by adding synonyms or training it on real questions and answers (Q&A setup is on Desktop's Modeling ribbon).

Quick insights

Source: Deck p.93

1. In the workspace click the **... menu** on the semantic model and choose **Quick insights**.
2. Wait for **Insights are ready**, then click **View insights**.
3. Review the AI-generated cards (up to 32); use the pin icon to pin one to a dashboard.

> **IMPORTANT**
>
> Quick insights are not available for reports or dashboards using roles (RLS) or DirectQuery.

Mobile layout

Source: Deck p.94-96

1. Open the report in **Edit** mode.
2. At the top switch from **Web layout** to **Mobile layout**.
3. Click **Auto-create mobile layout**, or drag selected visuals from the **Page visuals** pane onto the phone canvas and resize them.
4. Save the report.

> **IMPORTANT**
>
> You cannot build new content in Mobile layout. Build in Web layout first, then choose which visuals to arrange for phones.

The **Power BI mobile app** lets you interact with visuals, filters and Q&A; share links to workspace reports and dashboards with Pro users; draw and comment on screen (text, drawings, icons); and get notifications from your alerts or when you are @mentioned.

Part 10. Sharing and collaboration

The six ways to share

Source: Deck p.100-101

#	Method	Best for
1	Print or Export a report	Static PDF or PowerPoint copies (handouts).
2	Share reports or dashboards	Individual items when people only need to view and interact.
3	Share a workspace	Easiest teamwork: shared ownership and management of all content.
4	Publish an app	Large groups or teams; specific sets of datasets, reports, dashboards.
5	Embed securely online	Embedding in a site you own via SharePoint Online, URL or iframe.
6	Publish to web	Content that must be public to anyone online. Use with caution.

PRO TIP

When you share reports and dashboards, the maximum is 100 recipients at a time (500 in total). For more people, split into several sends or use user groups.

Workspace roles

Source: Deck p.103

Role	Can do
Viewer	Read-only: view workbooks, reports and dashboards. Cannot access datasets or dataflows or edit content. Used with row-level security.
Contributor	Publish, create, edit and delete content. Cannot add people to roles or modify members. Can share previously shared content but not new content.
Member	All Contributor rights plus: add members or users with lower permission levels, publish and update apps, share an item or app, allow others to reshare.
Admin	All Member rights plus: update/delete the workspace, add or remove other users (including admins).

Step-by-step: share a report

Source: Deck p.71

1. Open the report in Reading view and click **Share**.
2. In **Send link** choose who can use the link (default: people in your organization with the link can view and share).
3. Enter a name or email address, add an optional message and click **Send**. Or click **Copy link**, **Mail**, **Teams** or **PowerPoint** to send it another way.

Step-by-step: share the workspace

Source: Deck p.118

1. In the workspace header click **Manage access**.
2. Type the person's or group's name or email, pick a role from the drop-down (**Admin, Member, Contributor, Viewer**), then click **Add**.

Step-by-step: publish an app

Source: Deck p.104-105

1. In the workspace click **Create app** (it becomes **Update app** after the first publish).
2. **Setup tab:** enter the **App name** and **Description** (both required), add an **App logo** and pick an **App theme color**.
3. **Content tab:** click **+ Add content**, choose reports and dashboards from the workspace and set the **navigation order**.
4. **Audience tab:** choose **Entire organization** or **Specific users or groups** (type names/emails); the **Advanced** section holds extra options.
5. Click **Publish app**. The **Successfully published** box gives a link you can **Copy**, or click **Go to app**.

> **CHECK**
>
> After publishing, the app appears under **Apps** in the navigation pane and opens from the direct link. App menus let users print, share, export, explore data, comment, subscribe, favorite and (permission permitting) edit.

Step-by-step: publish to web (public)

Source: Deck p.106

1. Open the report. In Reading view choose **File > Embed report > Publish to web (public)** (in Editing view it is **File > Publish to web**).
2. In **Embed in a public website** click **Continue**.
3. Read the warning, then click **Publish** only if you accept that the report will be **100% accessible to anyone on the internet**.
4. The **Success** box gives a link and an HTML iframe embed code, plus **Size**, **Placeholder image** and **Default page** options.

> **IMPORTANT**
>
> Do not publish confidential or proprietary information or personal data. To manage or delete embed codes go to **Settings > Manage embed codes**.

Usage metrics report

Source: Deck p.107

1. Open the report and choose **... > Open usage metrics** (or the item's **... > View usage metrics report** in the workspace).
2. Make sure **New usage report on** is switched on.
3. Use the pages: **Report usage** (opens, views, viewers), **Report performance** (open time, browser, device), **Report list** (key metrics by report) and **FAQ**.

> **PRO TIP**
>
> Choose **Save report as** from the File menu to get at the underlying data and build a custom usage report.

Assignment 6: export to interactive PowerPoint

Source: Deck p.108-109

1. Open the **Restaurant Inspections** report in Reading view.
2. Click **Export > PowerPoint** and choose the **Embed live data** option.
3. Include both pages (**Inspections & Violations** and **Zip Code Detail**) and open the file in PowerPoint.
4. Look at the Zip Code Detail slide and note what differs from the report page.

VERIFY

The deck asks "what do you notice about the Zip Code Detail page?" but does not print the answer (its solution preview shows the first page). Compare the slide with the report page and note any visual or slicer that does not display.

Part 11. Row-level security (RLS)

Row-level security controls who can see specific data within a report based on rules you configure for individuals or groups. The cycle is **Create, Test, Deploy**.

> **PRO TIP**
>
> Object-level security (hiding columns or tables) is possible but needs third-party tools like Tabular Editor (deck p.112).

Static roles

Source: Deck p.114

A static role filters data out of the model for a specific audience using a simple DAX filter. It is not the same as a bookmark or a pre-filtered view. Static roles can be built in Desktop or Service but are only **implemented** in Service.

1. In Desktop open the **Modeling** tab and click **Manage roles**.
2. Click **+ New** under **Roles** and type the role name (deck example: *Los Angeles*).
3. Under **Select tables** click the table to filter (deck: **Region Lookup**).
4. In **Filter data** type the DAX expression, for example:
5. Click the **check mark** to validate the expression, then **Save**.

```
[facility_region] = "Los Angeles"
```

> **IMPORTANT**
>
> DAX used to define roles must be an expression that filters rows (returns TRUE/FALSE). It cannot reference a column, table or measure alone.

Test with View as

Source: Deck p.115

1. Open **Modeling > View as**.
2. Tick the role (the list also offers **None** and **Other user**) and click **OK**.
3. A yellow banner reads **Now viewing as: Los Angeles**. Check that the visuals only show that region.
4. Click **Stop viewing** when done.

Dynamic roles with USERPRINCIPALNAME

Source: Deck p.116-117

Dynamic roles filter by whoever is signed in, using one DAX function. It needs a table in the model that lists users: the deck's Inspector Lookup table has an **email_address** column (e.g. bfinnegan@maveninspectionalservices.onmicrosoft.com) and an **inspector_role** column (Inspector, Lead Inspector, Chief Inspector).

```
[email_address] = USERPRINCIPALNAME()
```

- Create the role (deck example: *Inspector*) on the Inspector Lookup table with that filter.
- `USERPRINCIPALNAME()` takes no parameters and returns the signed-in user's principal name (looks like an email address, e.g. aaron@mavendemo.com).
- Technically it is three parts: the username, the @ symbol and the company domain.

Applying RLS in Service

Source: Deck p.118

1. Publish the model. In the workspace click **Manage access** and confirm the people assigned to RLS roles have the **Viewer** role.
2. Click the semantic model's **... menu** and choose **Security**.
3. Select the role on the left (deck: *Los Angeles (1)*), type the **people or groups who belong to this role** and click **Add**.
4. Test by viewing as the role: the screen shows **Viewing: Restaurant Inspections - Now viewing as: Los Angeles**.

> **IMPORTANT**
>
> If a user is Admin, Member or Contributor, RLS will NOT work for them. Only users with the Viewer role are filtered.

Assignment 7: give David access to regions 1-8

Source: Deck p.119-120

1. In Desktop, **Modeling > Manage roles > + New**; name the role **Lead Inspector**.
2. Select the **Region Lookup** table and enter this filter, then validate and save:
3. **Test in Desktop** with **View as > Lead Inspector**. In Report view and Data view only regions 1-8 should show.
4. **Publish** to Power BI Service, then add David to the role under the semantic model's **Security** page and use the view-as-role option to confirm.

5 **Spot check:** the slicer on the **Zip Code Detail** page shows only Facility Regions 1 to 8.

```
[region_id] IN {1,2,3,4,5,6,7,8}
```

region_id	facility_region
1	Angeles Forest
2	Antelope Valley
3	Central L.A.
4	Eastside
5	Harbor
6	Los Angeles
7	Northeast L.A.
8	Northwest County

CHECK

Expected result (deck p.120): the slicer lists Select all, Angeles Forest, Antelope Valley, Central L.A., Eastside, Harbor, Los Angeles, Northeast L.A., Northwest County.

Part 12. Administration

Admin roles

Source: Deck p.124

Scope / admin role	Tasks
Microsoft O365 Global Admin	Unlimited access to all organizational features; assigns roles to other users.
Billing Admin	Manage subscriptions, purchase licences.
License Admin	Assign or remove licences for users.
User Admin	Create and manage users and groups, reset passwords.
Power Platform and Fabric administrator	Full access to Fabric management tasks; enable/disable Fabric features; report on usage and performance; review and manage auditing.
Power BI Premium (capacity) Admin	Assign workspaces to the capacity, manage user permission, manage workloads/memory, restart the capacity.

O365 admins manage Microsoft 365 services; Power Platform and Fabric admins have full access to Fabric management; capacity admins create, manage and restart Embedded capacity. Typical holders range from IT or a business owner to a director, manager or team lead. More: learn.microsoft.com/en-us/fabric/admin/microsoft-fabric-admin.

Admin portal

Source: Deck p.125

The admin portal menu lists: Tenant settings, Usage metrics, Users, Premium Per User, Audit logs, Domains, Capacity settings (Refresh summary), Embed Codes, Organizational visuals, Azure connections, Workspaces, Custom branding, Protection metrics, Featured content, Help + support.

- Configure tenant settings; monitor usage across the organization.
- Manage users and access audit logs (O365 admin center).
- Set capacity settings for Power BI Premium, Embedded, Trial and Fabric.
- Manage, view and edit embed codes, organizational visuals, workspaces and featured content.
- Apply custom branding; monitor data protection metrics and sensitivity labels (Purview and Defender).

> **IMPORTANT**
>
> Admin tools are not available to everyone. You may need to request access from an administrator in your organization.

Tenant settings

Source: Deck p.126

Tenant settings manage, configure, control and deploy features. Categories: Microsoft Fabric, Help and support, Domain management, Workspace, Information protection (extra licensing), Export and sharing, Discovery, App, Integration (XMLA endpoints, ArcGIS maps, SSO, OneLake), Power BI/R/Python visuals, Audit and usage, Dashboard, Developer, Admin API, Gen1 dataflow, Template app, Q&A, Semantic model security, Advanced networking, Metrics, User experience experiments, Insights, Datamart, Data model, Quick measure suggestions, Scale-out, OneLake, Git integration, Copilot and Azure OpenAI Service, and Preview/new/updated options.

> **IMPORTANT**
>
> Some tenant settings may expose your organization to risk. Always configure the tenant to match your organization's security policies and best practices.

Custom branding

Source: Deck p.127

- **Logo:** appears at the top-left of the navigation pane.
- **Cover image:** featured on the Power BI Service home screen.
- **Theme color:** background colour of the top navigation bar on all pages.

Help and support settings

Source: Deck p.128

Add custom URLs to the Power BI help menu so users reach internal resources: training documentation, discussion forums, licensing requests and the help/support desk.

Featured content

Source: Deck p.129

1. Open the report's **Settings** and turn on **Enable Feature on Home**.
2. Open Power BI **Home**; the report appears in the **Featured** section.

> **PRO TIP**
>
> Not seeing featured content? Switch the Home view from **Simplified** to **Expanded**.

Part 13. Master checklist: the whole project in order

Tick each line as you finish it. The part number tells you where the detail is.

Done	Area	Task	Part
[ ]	Account	Sign up with a work/school email or a Maven @mavenbi.org account; sign in at app.powerbi.com	3
[ ]	Account	Start the Fabric trial (profile icon > Start trial); header shows days left	3
[ ]	Workspace	Create the workspace: name, description, Advanced > Trial licence mode, Apply	4
[ ]	Workspace	Open Workspace settings; enable Users can edit data models in the Power BI service	4
[ ]	Data	Load Restaurant Inspection Data.csv via + New > Semantic model; answer the 3 profiling questions	5
[ ]	Gateway	Install the standard gateway; confirm status Online	6
[ ]	Dataflow	Create the Inspection Services dataflow (Text/CSV, gateway, Windows credentials), save, Refresh now	7
[ ]	Dataflow	Add the six lookup tables; Calendar date columns; promote headers; refresh again	7
[ ]	Schedule	Schedule dataflow refresh (e.g. 8:00 AM) and semantic model refresh (e.g. 8:30 AM)	8
[ ]	Model	In Desktop: connect to dataflow, build 6 relationships, facility_zip = Postal code, 12 measures	8
[ ]	Model	Save and Publish to the workspace	8
[ ]	Report	Build page 1 Inspections & Violations and page 2 Zip Code Detail; Save as Restaurant Inspections	9
[ ]	Dashboard	Pin Zip Code Detail as a live page and the 4 tiles; arrange tiles	9
[ ]	Alerts	Alert 1: grade below 92; Alert 2: inspections above 6500; at most every 24 hours	9
[ ]	Extras	Try Q&A and Quick insights; set up Mobile layout	9
[ ]	Sharing	Share the report, set workspace roles, publish an app, review usage metrics	10
[ ]	Sharing	Export to PowerPoint with Embed live data	10
[ ]	Security	Create Lead Inspector role ([region_id] IN {1,...,8}); test; publish; assign David as Viewer	11
[ ]	Admin	Review admin roles, tenant settings, branding and featured content (if you have admin rights)	12

Part 14. Troubleshooting and quick reference

Problems and fixes

Symptom	Likely cause and fix (from the deck)
Sign-up refuses your email	Power BI needs a work or school address. Use a Maven Analytics account (@mavenbi.org) instead (p.21-22).
No Fabric items after the trial	Microsoft changes trial options; you may not get Fabric items (p.22).
Cannot upload a file / browse in the dataflow	OneDrive for Business access is required (p.45).
Dataflow credentials rejected	Use the Windows username from `whoami` and your Microsoft account password if you sign in with PIN or biometrics (p.46-47).
Dataflow does not show in Desktop	You must refresh the dataflow first (p.49).
Gateway not working	Check Manage connections and gateways shows Online; click the status icon for tips; gateway PC must be on and running Windows (p.37, 40).
Scheduled refresh shows stale report	Dataflow and semantic model refresh at the same time. Offset them (p.59).
Excel scheduled refresh fails	The Excel file needs a data model (p.37).
Cannot pin a slicer	Pin the whole page as a live page instead (p.81-82).
Alert option missing	Only gauge, KPI and card tiles; not live pinned pages (p.89).
Quick insights unavailable	Not offered with RLS or DirectQuery (p.93).
Cannot build in Mobile layout	Build in Web layout first (p.94).
RLS does nothing for a user	User is Admin, Member or Contributor; assign Viewer (p.118).
Cannot send to many people	Limit is 100 recipients per send / 500 total; split or use groups (p.101).
Someone cannot view shared content	They need Pro/PPU or access to Premium/Fabric capacity (p.102).
Can't find Admin portal items	Admin tools need an administrator to grant access (p.125).

Numbers worth remembering

Item	Value
Trial length	60 days
Refreshes per day: Pro / PPU	8 / 48
Model size limit: Pro / PPU / Premium	1 GB / 100 GB / 400 GB
Quick insights cards	Up to 32
Share recipients	100 at a time, 500 total
Facilities in the dataset	44,031 (distinct)
Alert thresholds	Grade below 92; inspections above 6,500 per month
Dataset date range in the report slicer	7/1/2015 - 12/31/2017

Appendix A. What your .pbix file contains

The file **Powerbi_service_first_project.pbix** (about 55 KB) was opened and its internal parts were read. Here is exactly what was found:

Part	Finding
Origin	Metadata says it was created from the Cloud (Power BI Service), release 2025.01. File format version 1.28; saved 24 January 2025.
Tables	One table named **Violations (Demo)** (from the model diagram layout).
Report pages	One page, named **Page 1**, 1280 x 720, with **no visuals** on it.
Theme	Built-in theme CY24SU10 (a standard Microsoft theme).
Settings	Type detection and relationship import are switched on (Desktop defaults).
Data model contents	Compressed (XPress9). Column names, relationships and measures could not be read here.

VERIFY

Because the model is compressed, this guide cannot tell you what columns Violations (Demo) holds or whether it has measures. Open the file in Power BI Desktop and look at Model view and Data view to confirm.

How to use this file in the project

1. It is a starter you can publish to practice the Service steps: in Desktop open it and choose **Home > Publish**, or in Service use **Upload > Browse this device** (Part 5).
2. To build the full Maven Inspectional Services model, follow Parts 6 to 8 with the course data files; then rebuild or replace Violations (Demo) with the dataflow tables.
3. Keep the file name when republishing so the semantic model is replaced instead of duplicated.

Appendix B. Source map

Guide part	Deck pages
How to use / Part 1 project	2-7
Part 2 concepts	8-20, 29
Part 3 account and trial	21-23
Part 4 interface and workspace	24-28, 60
Part 5 connecting to data	30-36
Part 6 gateways	37-40
Part 7 dataflows	41-57
Part 8 refresh and semantic model	58-64
Part 9 reports and dashboards	65-97
Part 10 sharing	98-110
Part 11 row-level security	111-121
Part 12 administration	122-130

EXTRA

Items tagged EXTRA and the suggested DAX formulas are not in the deck. Everything else is taken from the deck pages cited.